EE5203 L09 Lab Card

Input capture — measure your own PWM - Basri Erdogan

Mission: Close the loop: TIM2 generates PWM on PA5 (your L08 project), one jumper wire carries it to PB6, and TIM4 captures it back. Measure the frequency and duty you configured—and prove they match.

Prepare before class

- ☐ Re-read the theory slides up to "The indirect trick."
- ☐ Bring your working L08 PWM project (TIM2 CH1 on PA5).
- ☐ Bring **one jumper wire**. On the Arduino header: PA5 = **D13**, PB6 = **D10**.
- ☐ Know the answer to: at a 1 µs tick, what period do `ARR = 999` captures show?

Learning objectives

By checkoff time, you can:

1. configure direct and indirect capture channels on TIM4;
2. turn hardware timestamps into period, frequency, and duty—wrap-safe;
3. verify a measurement against the generating side's own registers;
4. observe `CC1IF`, a live polarity flip, and an overcapture in the SFR view.

Hardware and files

Item	Required value
Board	Nucleo-F401RE
Wiring	one jumper: PA5 (D13) → PB6 (D10)
Generator	TIM2 CH1 PWM on PA5 — 1 kHz, from the L08 project
Analyzer	TIM4 CH1/CH2 capture on PB6, 1 µs tick, free-run

Configure in CubeMX

Starting from a copy of the L08 project, saved as `L09_Capture`:

1. `PB6` → `TIM4_CH1`.
2. **Timers → TIM4**: Channel 1 = `Input Capture direct mode`, polarity **rising**; Channel 2 = `Input Capture indirect mode`, polarity **falling**.
3. Prescaler `15` (1 µs tick), Counter Period `65535` (free-run).
4. **NVIC**: enable **TIM4 global interrupt**. Generate; verify — do not add.

Checkpoint A - Captures arrive

Start both sides:

```
HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);            // generator, 35 %
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 350);
HAL_TIM_IC_Start_IT(&htim4, TIM_CHANNEL_1);          // analyzer
HAL_TIM_IC_Start_IT(&htim4, TIM_CHANNEL_2);
```

Evidence for Checkpoint A: a breakpoint in `HAL_TIM_IC_CaptureCallback` is hit; in the SFR view `TIM4->CCR1` **changes by itself** between pauses—hardware is timestamping without you.

Checkpoint B - Measure the period

In the callback (CH1, rising edges):

```
uint16_t now = HAL_TIM_ReadCapturedValue(htim, TIM_CHANNEL_1);
period_us = (uint16_t)(now - last);     // wrap-safe on 16 bits
last = now;
```

Predict first: the generator runs `PSC = 15`, `ARR = 999` → what should `period_us` read? Then check the live variable.

Evidence for Checkpoint B: `period_us` ~ 1000; change the generator's `ARR` to `1999` and watch the measurement follow to ~ 2000 (and the LED frequency halve).

Checkpoint C - Measure the duty

CH2 (falling, indirect) gives the high time relative to the last rising edge:

```
high_us = (uint16_t)(HAL_TIM_ReadCapturedValue(htim, TIM_CHANNEL_2) - last);
```

Duty = `high_us / period_us`. With the generator at `CCR1 = 350`, predict both numbers before looking.

Evidence for Checkpoint C: measured duty ~ 35 %; run the L08 breathing sweep on the generator and watch the measured duty track it live.

Individual extension

Pick one, predict first:

- **Overcapture:** pause in the debugger through several PWM periods, resume, and find `CC1OF` set in `TIM4->SR`. Explain what was lost.
- **Polarity flip:** clear/set `CC1P` in `TIM4->CCER` from the SFR view and explain what the period measurement does (and does not) change.
- **Ultrasonic (hardware permitting):** HC-SR04 echo width on CH1/CH2; `distance_cm = width_us / 58`.

Acceptance checklist

- ☐ `.ioc`: TIM4 CH1 direct/rising, CH2 indirect/falling, PSC 15, free-run.
- ☐ Callback checks instance **and** channel; subtractions are `uint16_t`.
- ☐ Measured period and duty match the generator's `ARR`/`CCR` numbers.
- ☐ Measurement tracks a live change on the generating side.
- ☐ One extension demonstrated with the SFR evidence.

Individual oral check

- Who writes `TIM4->CCR1`, and at what instant exactly?
- Why must the subtraction be cast to `uint16_t`?
- Why is CH2 configured *indirect*? What would *direct* listen to?
- Your period reads 3036 after captures 64000 and 1500—explain.
- What is the slowest signal this 16-bit setup can measure honestly?

Submit

Submit `main.c` and one SFR-view screenshot showing `TIM4->CCR1`, `CCR2`, and `SR` with a capture flag set—plus your predicted-vs-measured table for Checkpoints B and C.

If you are stuck

Walk the chain: jumper seated (D13 → D10)? Generator actually running (LED dim/bright)? `PB6` in AF2 in the `.ioc`? `Start_IT` called for **both** channels? Breakpoint in the callback reached? Then: **period garbage** — check the cast and the polarity; **duty > 100 %** — CH2 is probably direct, or the edges are swapped; **CCR1 frozen** — capture never started or the NVIC box is unticked.